

Diesel Engines
V 4000 R 41L
V 4000 R 42

Maintenance Schedule
M050605/02E

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

The Maintenance Schedule is based on the power output defined in the individual orders, under consideration of the operating parameters established in the order's specifications.

The time intervals according to which the maintenance tasks and the relevant checks and tasks involved are to be completed are based on operational experience and can only serve as guidelines. Specific operational conditions may require modification of the Maintenance Schedule.

Note:

The engine manufacturer cannot be held responsible for the performance of the following maintenance tasks: The operator is responsible for carrying out the following maintenance tasks:

Fuel filter

The maintenance interval depends on the degree of fuel contamination. The fuel filter must be replaced after 2 years at the latest.

Battery

Battery maintenance depends on the degree of use and ambient temperature. The battery has to be checked according to the manufacturer's instructions, as it is not always a part of the MTU scope of supply.

Fluids and Lubricants:

Only those fluids and lubricants are to be used that are approved in the MTU Fluids and Lubricants Specification A001061. A special note must be made that, after maintenance or repair work on the cooling system, the operator must check the concentration of the corrosion inhibitor or the corrosion-inhibitor antifreeze.

Major Overhaul

Requires complete engine disassembly to facilitate repair/overhaul of all relevant components.

Out-of-Service Periods

If the engine is to remain out of service for several weeks, spray engine oil into the air-intake ducting. Actuate the engine shutdown system and bar the engine with the starting equipment when performing this task. Carry out maintenance tasks in accordance with the time limit schedule (see the Maintenance Frequency Chart). In addition, the engine must be barred manually every month. If the engine is to remain out of service for more than 1 month, carry out engine preservation in accordance MTU Fluids and Lubricants Specification, Publication No. A001061.

Application group

2A: Rail Traction

Maintenance Frequency Chart

Maintenance Level	W1: Operational checks, daily	X
	W2: Operating hours	500
	Limit, months	6
	W3: Operating hours	1 000
	Limit, year	1
	W4: Operating hours	4 000
	Limit, year(s)	2
	W5: Operating hours	8 000
	Limit, year(s)	6
	W6: Operating hours	24 000
	Limit, year(s)	18

Maintenance Task Schedule

	Maintenance Level W1 – Operational checks	
G000.000.13	Engine operation	Check speed (if gauges are provided)
G000.000.15	Engine operation	Check pressures (if gauges are provided)
G000.000.17	Engine operation	Check temperatures (if gauges are provided)
G000.000.19	Engine operation	Check running noises
G000.000.21	Engine operation	Check visually for leaks and general condition of the engine
G083.101.03	Fuel pre-filter	Drain water and contaminants
G111.071.01	Intercooler	Check drain line for water discharge and obstructions
G121.052.01	Air filter	Check service indicator
G140.000.05	Exhaust system	Check exhaust gas color
G180.000.01	Engine oil	Check level
G202.000.03	Engine coolant	Check level
G202.051.01	Coolant pump	Check relief bore for obstructions
G203.051.01	Charge air coolant pump	Check relief bore for obstructions

	Maintenance Levels	W	2	3	4	5	6
G180.000.04	Engine oil	Take sample and analyze					
G180.000.05	Engine oil	Change - With oil category 1, with centrifugal oil filter Every 500 operating hours - With oil category 2, with centrifugal oil filter Every 1000 operating hours - With oil category 3, with centrifugal oil filter Every 1500 operating hours After 2 years at the latest					
G183.052.03	Engine oil filter	Replace Accomplish when changing the oil					
G183.101.01	Centrifugal oil filter	Clean, check thickness of oil sludge layer Replace paper sleeve					
G055.050.01	Valve gear	Check valve clearance					
G083.051.03	Fuel duplex filter	Change throw-away filter					
G083.101.09	Fuel pre-filter	Replace filter element and sealing ring					
G121.051.03	Air filter	Clean					
---	Air ducting	Check ducting between air filter and turbocharger visually for leaks and damage					
G205.051.01	Coolant cooler	Check cooler elements for external contamination					
G205.051.03	Charge-air coolant cooler	Check cooler elements for external contamination					
G018.101.02	Crankcase ventilation	Replace filter elements					
G073.051.04	HP fuel delivery pump	Replace Every 2nd W4 maintenance level = 12000 operating hours					
G075.051.05	Fuel injector(s)	Replace					
G121.051.01	Air filter	Replace filter elements					
G175.100.01	Flame-torch starting system	Replace flame torch (if provided)					
G202.000.05	Engine coolant	With water-soluble corrosion inhibitor: Take sample and analyze Change coolant, see MTU Fluids and Lubricants Specification					
G213.051.01	Charging generator	Check condition of coupling, replace if necessary					
G223.052.01	Hydraulic pump drive(s) (Equipment carrier / Crankshaft)	Check condition of coupling(s), replace if necessary					
G252.051.03	Coupling (Küsel)	Check radial clearance					
---	Electrical preheating unit	Check pump for leaks, replace rotary seal if necessary, motor (visual inspection) (if provided)					
	Prior to commencing W5 maintenance, run the engine and record all measured values, then drain engine coolant and flush cooling system!						
	Valve gear	Check rocker arms, valve bridges, swing followers and swing follower bearings for wear					
	Cylinder heads	Replace					
	Running gear	Check piston crowns (visual inspection)					
	Crankcase	Check cylinder liner wear pattern					
	Vibration damper	Replace					
	Turbocharger	Replace					
	Air ducting	Clean					
	Intercooler	Replace					
	Starter	Replace					
	Charging generator	Replace					
	HP fuel delivery-pump sensor	Replace					
	Engine coolant	With antifreeze: Take sample and analyze Change coolant, see MTU Fluids and Lubricants Specification					
	Coolant pump	Replace					
	Charge air coolant pump	Replace					
	Coolant cooler	Clean and leak test					
	Charge-air coolant cooler	Clean and leak test					
	Electrical preheating unit	Maintenance is the same as for the engine (if provided) Carry out visual and acoustic check of the three-phase motor, replace antifriction bearing(s) if necessary Re-grease motor's antifriction bearing(s)					
	Engine oil heat exchanger	Remove, clean, leak test and replace seals					
	Engine mounts	Check set adjustment of the central buffer					

	Hydraulic pump drive(s) (Equipment carrier / Crankshaft)	Replace coupling(s)	
	Electrical fuel delivery pump	Pump operates for a maximum of 1% of the engine operating time D.C. motor – Check length of carbon brushes and condition of commutator – Remove carbon residue from engine interior	(if provided)
	Centrifugal oil filter	Check, replace bearings if necessary	
	Coupling (Küsel)	Overhaul	
	Major Overhaul	Total disassembly of the engine; inspection / repair of all components	